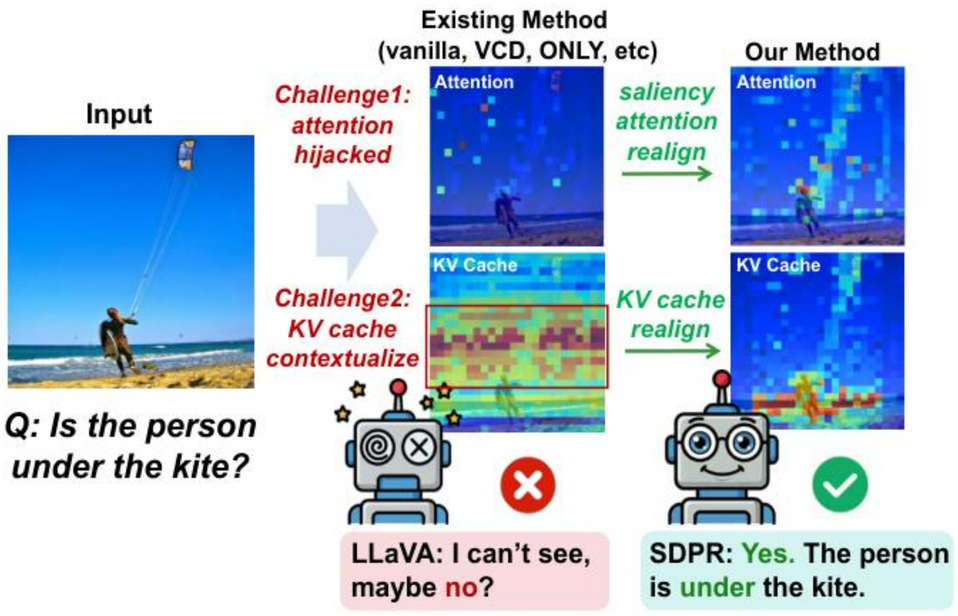
Look Clearly Before Answering: Mitigating Hallucinations in LVLMs via Saliency-Driven Perceptual Realignment

Pengxu Chen[1], Yao Zhu[2,†], Guangming Zhu[1], Jun Sheng[3], Jincai Huang[4], Xiangyang Ji[2], Liang Zhang[1,†]

[1] Xidian University, [2] Tsinghua University, [3] Shanghai Road Transport Development Center,
[4] Hunan Institute of Advanced Technology ,[†]corresponding authors

ACM Multimedia 2026
Rio de Janeiro, Brazil

Background & Motivation



LVLMs hallucinate: fluent but inconsistent with visual evidence.

Existing hallucinations mitigation methods: only target language prior bias or cross-modal imbalance.

Our finding: visual awareness progressively degrades across *perception→memory→decoding*.

Failures Patterns:

- **attention hijacking** by non-semantic sink tokens.
- **KV-cache asymmetric** contextualization.

GOAL: mitigates the degradation of visual awareness and suppresses hallcinations.

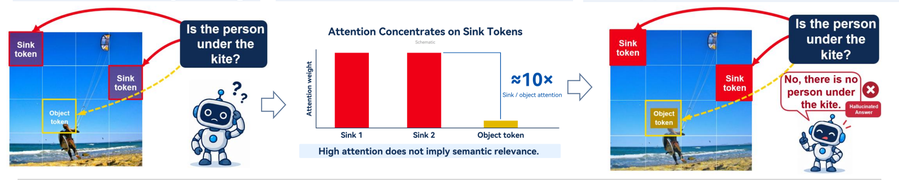
Challenges & Contributions

Why Do LVLMs Hallucinate?

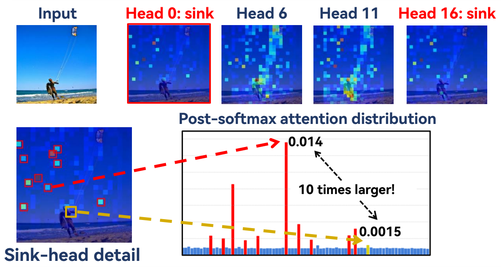
Progressive degradation of visual awareness in perception and memory.

Perception problem: attention is hijacked by non-semantic sinks

1 Attention Hijacking | 2 Concentrated Attention | 3 Suppressed Visual Evidence



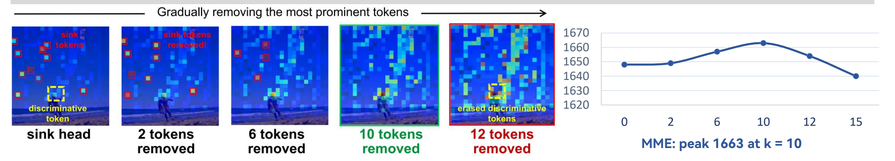
(a) Attention Sink Head: where the attention goes?



Attention concentration

Metric	Faithful	Hallucinatory	H / F
Gini	2.85%	4.88%	1.71×
Top-1	2.40%	4.29%	1.79×
Top-5	10.79%	19.32%	1.79×
Top-10	20.09%	35.82%	1.78×

(b) Removing top-k tokens reveals evidence, but has a trade-off

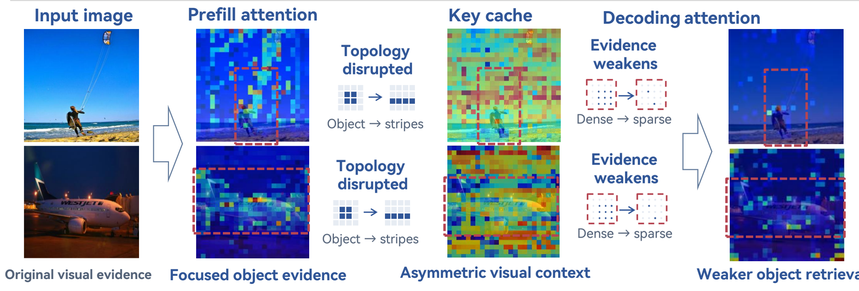


Core problem: non-semantic sinks overwhelm discriminative visual evidence without erasing it.

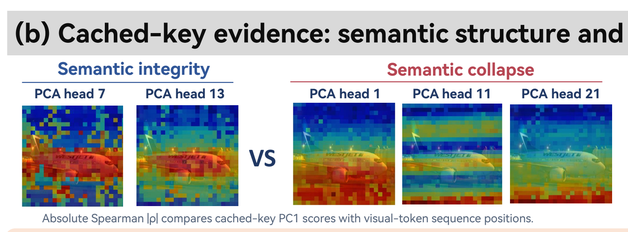
Our Solution: Saliency-Driven Attention Redistribution
Redirect attention from sinks while preserving discriminative evidence.

Memory Problem: visual topology distorts and weak retrieval

(a) Visual Cache Distortion: why the model can't memory?



(b) Cached-key evidence: semantic structure and sequence-position



Absolute Spearman |ρ| ↓ (MME, paper Table 2)

Faithful	0.0178
Hallucinatory	0.0289

1.62× stronger sequence-position dependence

Absolute Spearman |ρ| compares cached-key PC1 scores with visual-token sequence positions.

Core problem: causal masking disrupts visual topology and weaken object-relevant retrieval.

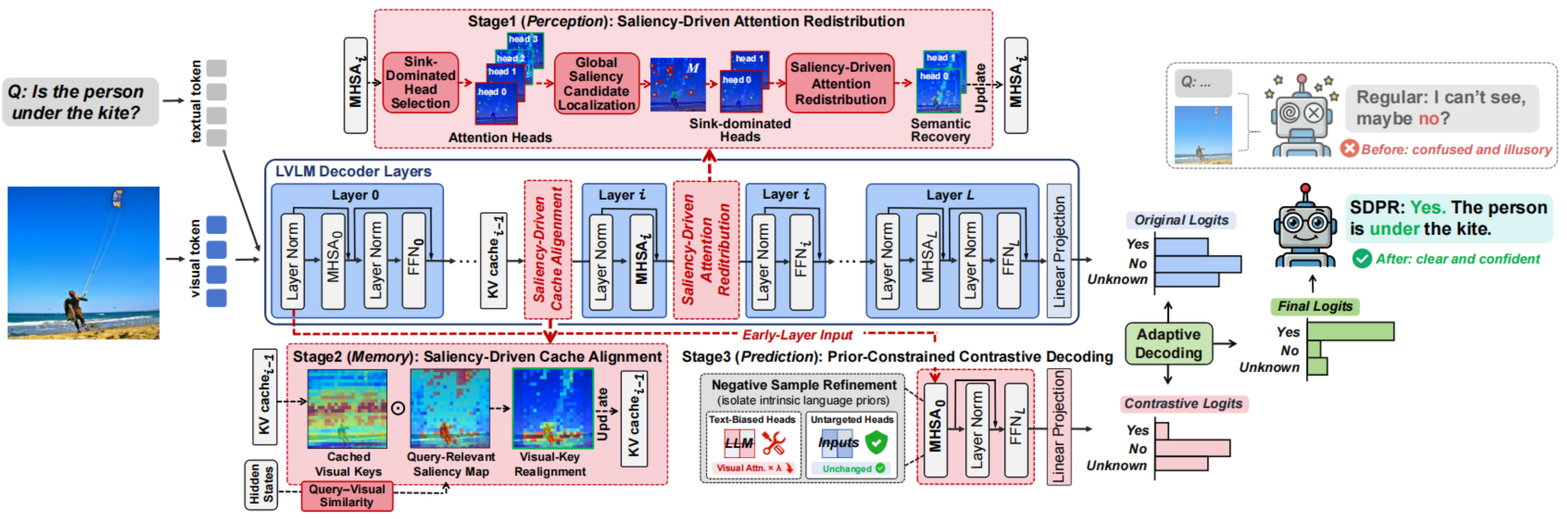
Our Solution: Saliency-Driven Cache Alignment
Counteract position-dependent distortion to preserve query-relevant memory.

Causal Chain: how does the model generate illusions?

None-semantic Attention Sinks → Salient visual Evidence Suppressed → Position-Dependent KV Cache Distortion → Visual Retrieval Weakened → Language Prior Dominates → **Hallucination**

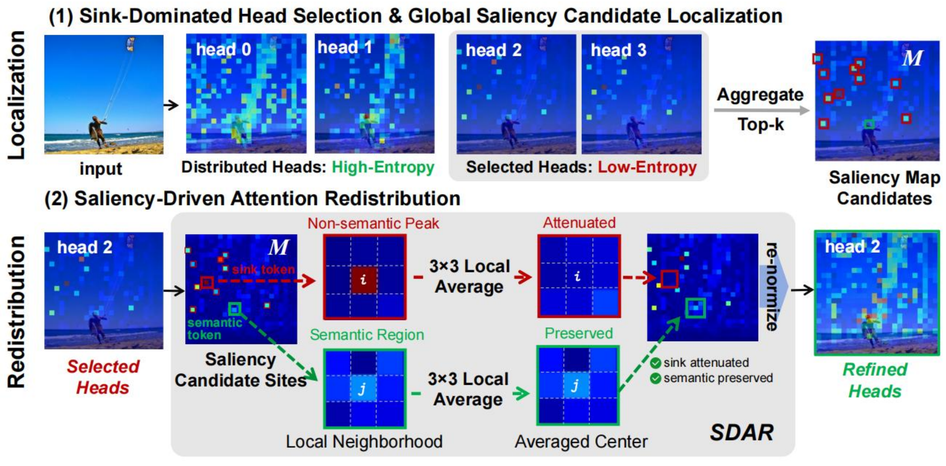
Method: Saliency-Driven Perceptual Realignment

SDPR: a training-free framework that mitigates the degradation of visual awareness and suppresses hallucinations.



① SDAR: Saliency-Driven Attention Redistribution

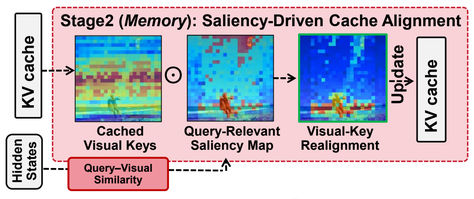
Localize candidates, then refine each head using spatial context



② SDCA: Saliency-Driven Cache Alignment

Update visual keys; keep other keys and all values unchanged.

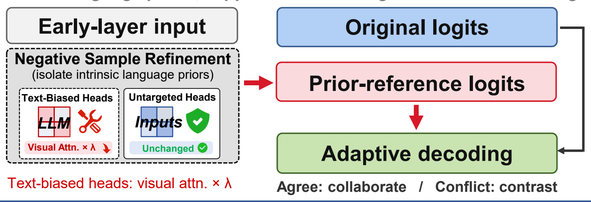
Query + visual hidden states → Similarity → S_q



$$K_V = K_V \odot S_q$$

③ PCD: Prior-Constrained Contrastive Decoding

Estimate language priors, suppress them through contrastive decoding.



Practical Performance

Evaluations

Hallucination reduction, generalization and efficiency

+3.68 pp
POPE F1 · COCO average · LLaVA-1.5

21.2 → 15.2
CHAIR C_S ↓ · 64 tokens · LLaVA-1.5

1648.0 → 1855.9
MME total ↑ · LLaVA-1.5

1 Hallucination benchmarks

POPE: dataset averages / LLaVA-1.5

Method	COCO AVG Acc ↑	COCO AVG F1 ↑	A-OKVQA AVG Acc ↑	A-OKVQA AVG F1 ↑	GQA AVG Acc ↑	GQA AVG F1 ↑
Baseline	80.900	82.266	74.844	78.866	74.011	78.400
VCD *CVPR24*	81.278	82.629	75.522	78.769	74.366	78.435
M3ID *CVPR24*	82.244	83.391	75.278	79.292	74.600	79.028
VAR *ICLR25*	80.633	81.873	74.787	78.893	74.022	78.400
VAF *CVPR25*	81.088	82.677	72.955	77.919	72.033	77.454
ONLY *ICCV25*	84.822	85.667	78.011	81.422	76.544	80.693
AvisC *ACL25*	78.250	80.581	71.767	77.419	70.167	76.633
MoD *ACL25*	84.544	85.252	77.456	81.099	75.822	80.220
SDPR *ours*	**85.233**	**85.946**	**78.799**	**81.923**	**76.922**	**80.867**

CHAIR: 64-token generation

Method	LLaVA C_S ↓	LLaVA C_I ↓	Qwen C_S ↓	Qwen C_I ↓	InternVL C_S ↓	InternVL C_I ↓
Baseline	21.2	7.4	31.6	10.3	15.4	6.8
VCD *CVPR24*	21.4	7.5	33.0	10.4	16.4	6.0
M3ID *CVPR24*	22.4	6.9	33.2	10.6	14.8	5.6
VAR *ICLR25*	22.2	7.6	32.6	10.0	20.2	8.6
VAF *CVPR25*	23.6	8.9	30.4	10.4	15.1	7.3
ONLY *ICCV25*	17.2	5.3	27.3	8.4	18.6	7.1
AvisC *ACL25*	20.6	6.5	32.2	10.0	16.1	7.2
MoD *ACL25*	20.6	6.4	30.4	10.1	15.4	6.8
SDPR *ours*	**15.2**	**4.7**	**24.2**	**7.6**	**14.6**	**5.3**

MME: total scores

Method	LLaVA 1.5-7B ↑	Qwen2.5 VL-7B ↑	InternVL 2.5-8B ↑
Baseline	1648.0	1827.0	1779.7
VCD *CVPR24*	1673.2	1881.3	1859.1
M3ID *CVPR24*	1680.3	1862.2	1918.0
VAR *ICLR25*	1624.6	1765.7	1803.1
VAF *CVPR25*	1675.8	1779.9	1867.2
ONLY *ICCV25*	1789.2	1900.6	1945.0
AvisC *ACL25*	1780.5	1899.4	2016.9
MoD *ACL25*	1744.1	1920.6	2022.4
SDPR *ours*	**1855.9**	**1938.1**	**2065.6**

2 General-purpose benchmarks

Overall average: 38.4 → 42.7

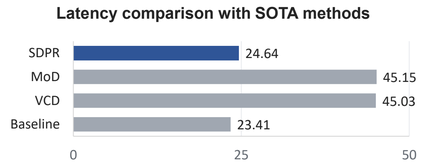
Method	LLaVA-1.5-7B MME-RW↑	MMVP↑	MMMU↑	MMMU-Pro↑	SQA-img↑	Qwen2.5-VL-7B MME-RW↑	MMVP↑	MMMU↑	MMMU-Pro↑	SQA-img↑	InternVL2.5-8B MME-RW↑	MMVP↑	MMMU↑	MMMU-Pro↑	SQA-img↑	Avg. ↑
Baseline	30.4	27.3	26.1	25.3	65.9	22.6	**29.3**	25.1	22.1	60.2	31.5	43.0	33.7	36.7	96.7	38.4
VCD *CVPR24*	34.0	29.3	25.2	26.2	65.2	23.7	24.0	26.8	23.3	62.0	32.8	39.3	36.1	38.0	97.0	38.9
M3ID *CVPR24*	33.2	**30.0**	26.1	26.5	65.5	24.4	22.0	27.8	23.6	63.4	32.3	43.3	40.7	39.4	97.1	39.6
VAR *ICLR25*	28.7	29.3	25.2	24.4	64.0	19.5	22.7	9.2	22.8	58.8	34.1	42.0	41.3	35.7	97.2	37.0
VAF *CVPR25*	33.4	26.6	26.1	25.0	65.4	23.6	22.7	27.6	23.5	61.8	32.7	40.0	34.6	37.1	96.7	38.5
ONLY *ICCV25*	33.5	28.7	28.0	26.1	67.1	25.0	28.0	32.4	26.9	**67.9**	32.5	43.3	35.2	40.1	97.3	40.6
AvisC *ACL25*	33.9	22.0	27.0	25.4	65.1	23.1	28.0	31.8	24.6	67.0	35.0	42.0	40.6	39.4	97.3	40.1
MoD *ACL25*	34.5	26.0	25.2	26.8	65.2	24.0	28.7	33.8	24.6	67.1	35.3	38.0	41.8	**40.5**	97.5	40.7
SDPR *ours*	**34.6**	29.3	**31.0**	**27.2**	**67.6**	**26.1**	**29.3**	**36.2**	**27.0**	67.5	**37.8**	**46.0**	**43.0**	40.2	**97.7**	**42.7**

3 Larger-scale model — LLaVA-1.5-13B

Method	POPE ↑	MME ↑	MME-RW ↑	MMVP ↑	MMMU ↑	SQA-img ↑
Baseline	79.2	1652	33.35	32.7	24.8	69.4
VCD *CVPR24*	81.4	1690	34.24	34.0	28.6	70.0
M3ID *CVPR24*	81.6	1720	33.45	33.3	28.1	70.0
VAR *ICLR25*	80.1	1711	30.38	28.7	27.3	67.4
VAF *CVPR25*	80.3	1716	30.34	28.1	26.5	67.9
ONLY *ICCV25*	84.9	1812	37.10	33.3	29.4	69.6
AvisC *ACL25*	84.3	1804	36.52	31.3	28.5	69.5
MoD *ACL25*	84.6	**1844**	36.91	**35.3**	30.5	70.0
SDPR *ours*	**85.2**	1834	**37.42**	34.0	**32.3**	**71.2**

4 Efficiency / 7B model

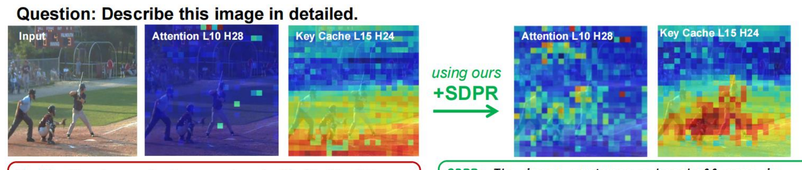
Method	GPU Mem ↓	Time (ms) ↓
Baseline	**16,820**	**23.41**
VCD *CVPR24*	17,186	45.03
M3ID *CVPR24*	17,186	44.03
VAR *ICLR25*	16,825	24.05
VAF *CVPR25*	16,825	**23.41**
ONLY *ICCV25*	16,840	25.37
AvisC *ACL25*	19,036	27.00
MoD *ACL25*	19,036	45.15
SDPR *ours*	16,840	24.64



SDPR: only +5.3% latency; +0.12% memory (54.71% latency compared with VCD)

Evaluations & Demo

Question: Describe this image in detailed.



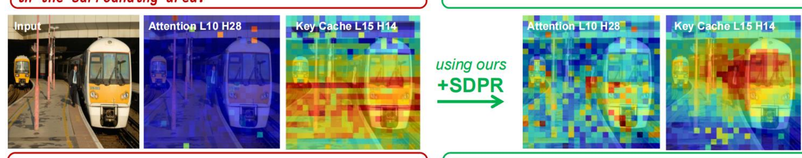
LLaVA: The image features a baseball field with several players on the field. A baseball batter is at the home plate, holding a bat and standing ready for the pitch. Behind him, players are positioned in various benches and stands, while others are scattered around the field.

SDPR: The image captures a baseball game in progress, with a batter standing at home plate, holding a baseball bat and preparing to swing. The catcher and umpire are also present, ready to catch the ball if the batter misses. There are several other people in the scene, likely team of other players.



LLaVA: The image features a large, green city public commuter bus driving down the street, either near a curb or a bus stop station. The bus appears to be picking up or dropping off passengers. Other vehicles, including a car and a truck, can be seen in the surrounding area.

SDPR: The image features a large green and white bus parked on the side of the road, likely at a bus stop. The bus is positioned in front of a building, possibly a bus stop or a nearby establishment.



LLaVA: The image features a train station with train tracks and a white passenger train sitting on the right side of the platform. A man wearing a suit and walking stick is gesturing to the train, he is carrying an umbrella with him and seems to be busily

SDPR: The image features a train station with a train parked on the tracks. A man is walking towards the train, likely preparing to board it. The train is positioned on the right side of the image, occupying a significant portion of the scene.

SDPR: Revived Perception ✅ · Robust Memory ✅ · Faithful Prediction ✅
➡ mitigating multimodal hallucinations for LVLMs